

Flying Your Imagination: Integrating AI in VR for Kite Heritage

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Figure 1: Kite-Flying Simulation, based on the Default Model in Spline and a 3D Kite.

Abstract

“Flying Your Imagination” investigates the innovative integration of Virtual Reality (VR), Artificial Intelligence (AI) technology, and embodied interaction design to revitalize and preserve China’s significant intangible cultural heritage (ICH) of kite culture. We designed a VR kite simulation environment that allows users to immerse themselves in the joy of kite flying and making within a virtual space. The system incorporates AI to generate visual designs for kites and offers diverse interactive scenarios, opening up new avenues for transmitting cultural heritage.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **User centered design**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

Digital Heritage, Virtual Environment, Artificial Intelligence

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1 Introduction

As an essential part of China’s intangible cultural heritage (ICH), kite culture faces significant preservation challenges due to modernization and urbanization. Urbanization has reduced kite-flying frequency; the urban green space in major Chinese cities has decreased by approximately 30% over recent decades [Song et al. 2020], and suitable open spaces are becoming increasingly scarce. Flying kites poses safety risks as well; during the Basant festival, kite-flying-related injuries accounted for 7.09% of all hospital admissions [Quddus 2010].

Virtual reality (VR) and artificial intelligence (AI) offer innovative ways to revitalize kite heritage digitally. VR simulates natural environments, allowing users to engage in kite-flying safely, and AI extends imagination by generating digital content. The embodied interaction design significantly enhances the VR experience by making it immersive and educational, as it emphasizes the user’s body and physical movements in interaction [Shen and Ghani 2023] and creates a dynamic unity of cognition, body, and environment. “Flying Your Imagination” aims to achieve an embodied interaction design combining VR and AI to create an engaging and educational kite culture experience, helping protect and promote this valuable heritage in contemporary environments.

2 Related Works

We analyzed previous works that applied VR and AI technologies in cultural heritage conservation, which inspired us to combine VR and AI to create an immersive and educational interactive kite heritage experience. VR reconstructs and reproduces the ecological context of the “Nvshu (Chinese women’s script)” culture, augmenting users’ sense of presence and immersion [Zhang et al. 2023]. Tsinghua University Shenzhen used AI to generate 3D models for reconstructing the historical context and details of the Ming Dynasty, providing an accurate and immersive representation of this historical period [He and Sun 2023]. Additionally, embodied interaction in VR enables learners to deeply understand and create



Figure 2: Different views of other 3D generated kites.

high-quality artworks related to intangible cultural heritage by integrating physical interaction with immersive virtual environments [Ji et al. 2021].

3 Technical Implementation

Our work demonstrates how VR and AI technologies can be leveraged to preserve and promote the intangible cultural heritage of Chinese kites. We have integrated an AI model into the VR environment using advanced text-to-image and image-to-3D model generation techniques. Specifically, we first generate high-resolution 2D visual references using DALL-E 3 and then employ the Tripo framework, a transformer-based architecture, to convert these 2D images into intricate 3D model meshes. Subsequently, these 3D models are further refined in Blender through mesh simplification, fragment removal, and mirroring techniques to ensure optimal visual quality and interactive performance in the VR environment. The figure showcases the diversity of designs (see Figure 2).

4 Project Description

“Flying Your Imagination” provides a dynamic unity composed of cognition, body, and environment through the Oculus Quest2, providing users with an educational and imaginary experience. We designed three interaction phases in our work: Kite Learning, Kite Making, and Kite Flying.

Kite Learning (traditional cognition) aims to enhance users’ engagement by helping them understand and experience the crafting process, divided into tool introduction, material selection, and handmade processes full of knowledge and production experiences. Users can immerse themselves in kite construction’s craftsmanship and cultural significance, which follows a regular routine designed to educate users and help them deeply understand this heritage.

Kite Making (imagination cognition) is a way to create personalized kite designs, where users can apply AI by typing their prompts in text-to-3D or image-to-3D for VR kite flying. They can adjust their designs in different styles. This embodied creating process can stimulate users’ imagination and let them realize that they could furthermore push away the boundaries of traditional heritage (see Figure 3).

Kite Flying (mental recognition) aims to offer embodied interaction to realistically replicate the physical actions and sensory experiences of kite flying (see Figure 1). We plan to add authentic feedback on the tension and pull of the kite string, capturing the resistance and subtle movements needed to control the kite under different wind conditions. The system offers various environments, including serene parks, vibrant beaches, refreshing polar ice caps, and fully customizable scenes with adjustable wind speeds, ambient sounds, and exquisite visual details to enhance immersion. This will create an interactive kite-flying adventure, making users feel like

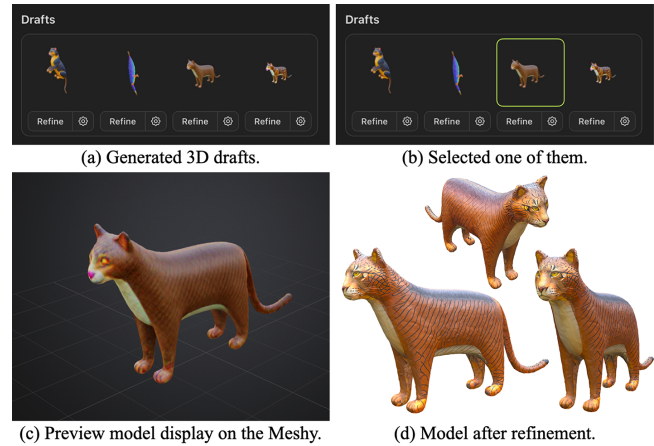


Figure 3: 3D Kite Making with Meshy.ai: We now use DALL-E 3, the Tripo framework, and Blender.

they are truly flying a kite in nature and achieving an immersive experience that embeds the body into the environment.

5 Conclusion and Future Works

From August to October 2024, we conducted two phases of informal user testing on three core functions: learning about kite history and culture, interactive production experiences, and AI-generated creations. The testing involved eight participants aged between 18 and 50, with an equal gender distribution. We collected feedback through questionnaires, observations, and semi-structured interviews. The results indicated good usability, though there remains room for improvement. Looking ahead, our future work will focus on two key aspects. First, we will enhance the realism of physical interactions, particularly by improving tactile feedback between the user’s hands and the kite string during the flying experience. After the final phase, we will conduct a formal user study by applying for approval from the Institutional Review Board (IRB). Second, we plan to expand the diversity of environmental scenarios to simulate kite flying under various wind conditions, such as on beaches, highlands, and snowy mountains. Through continuous innovation and iteration, we strive to develop a valuable tool for preserving and promoting traditional cultural heritage in modern contexts.

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